

Problem 5141

Expand the inner function $g(x)$:

$$g(x) = x(x + 2b)(x - 2b) = x^3 - 4b^2x$$

Find the local extrema by taking the derivative (find out how many times $g(x)$ hits a certain value y):

$$g'(x) = 3x^2 - 4b^2$$

Set $g'(x) = 0$ for the critical points:

$$x = \pm \frac{2b}{\sqrt{3}}$$

Evaluate $g(x)$ at these critical points:

$$M = \left(-\frac{2b}{\sqrt{3}}\right)^3 - 4b^2\left(-\frac{2b}{\sqrt{3}}\right) = -\frac{8b^3}{3\sqrt{3}} + \frac{8b^3}{\sqrt{3}} = \frac{16b^3}{3\sqrt{3}}$$

At $x = -\frac{2b}{\sqrt{3}}$ there is a local maximum. At $x = \frac{2b}{\sqrt{3}}$ there is a local minimum:

$$-M = -\frac{16b^3}{3\sqrt{3}}$$

Theorem. For any horizontal line $y = c$: 1. If $|c| < M$, the equation $g(x) = c$ has three real solutions. 2. If $|c| = M$, the equation $g(x) = c$ has 2 real solutions. 3. If $|c| > M$, the equation $g(x) = c$ has 1 real solution.

Let $y = g(x)$. The given equation is $|f(y)| = k$. Substitute $f(y) = (y + a)(y - a) = y^2 - a^2$ into the equation:

$$|y^2 - a^2| = k$$

This means

$$y^2 = a^2 + k \quad \text{or} \quad y^2 = a^2 - k$$

Because $g(x)$ is an odd function, the roots for y will come in positive and negative pairs $(\pm y_1, \pm y_2, \dots, \pm y_n)$. Consequently, the number of solutions for $+y_i$ will be the same as the number of solutions for $-y_i$. This means that non-zero pairs of y will always generate an **even** total number of solutions for x .

However, we are given that there are exactly 7 distinct real solutions. To get an odd number of total solutions, one of the values of y must be 0:

$$a^2 - k = 0 \implies k = a^2$$

Substituting $k = a^2$ into equations for y^2 :

$$y^2 = a^2 - a^2 = 0 \implies y = 0$$

$$y^2 = a^2 + a^2 = 2a^2 \implies y = \pm a\sqrt{2}$$

- For $y = 0$: $g(x) = 0 \implies x^3 - 4b^2x = 0$. This gives roots $x = 0, \pm 2b$, which is **3** solutions.

Since we need a total of 7 solutions, the remaining y values ($y = a\sqrt{2}$ and $y = -a\sqrt{2}$) must provide the other **4** solutions. Because of symmetry, they must each provide exactly **2** solutions.

As established in Step 1, $g(x) = c$ has exactly 2 solutions if and only if c is a local extremum.

Therefore, $a\sqrt{2}$ must equal the local maximum M :

$$a\sqrt{2} = \frac{16b^3}{3\sqrt{3}}$$

We need to find the value of $\frac{b^3}{a}$. Rearranging the equation above:

$$\begin{aligned}\frac{b^3}{a} &= \frac{\sqrt{2} \cdot 3\sqrt{3}}{16} \\ \frac{b^3}{a} &= \frac{3\sqrt{6}}{16}\end{aligned}$$

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